

Dynamics of Financing Frictions

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PhD Lectures in Corporate Finance

When do constrained firms escape the trap?

Outline

Setup

Self-financing convergence

Entry and exit

Stochastic productivity

Adjustment costs

Irreversibility

Risk aversion

Resolution

A constrained firm is stuck at suboptimal scale

Entrepreneur with productive project and limited collateral. Borrowing is capped at λ times net worth a :

$$k \leq \lambda a$$

Her actual capital k falls short of the first-best k^* . She earns MRPK above the user cost $r + \delta_k$, but cannot scale up

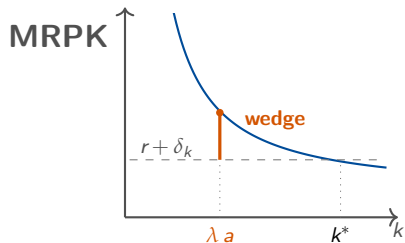
A constrained firm is stuck at suboptimal scale

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The question for the rest of the lecture: is this wedge a permanent trap, or can she save her way out?



The static model predicts a permanent wedge

Reduced-form revenue (DRS in k): $R(k, z) = z \cdot k^\alpha$, with $\alpha = 0.62 < 1$

First-best capital (from $\text{MRPK} = r + \delta_k$):

$$k^*(z) = \left(\frac{\alpha z}{r + \delta_k} \right)^{\frac{1}{1-\alpha}}$$

Actual capital: $k = \min(k^*(z), \lambda a)$

Constrained firms ($k < k^*$) have $\text{MRPK} > r + \delta_k$

→ Static model: wedge is **permanent**

→ But static model ignores **accumulation**

Primitives pin down user cost, returns, and leverage

Symbol	Meaning
r	Return on the safe asset, the return on household saving
δ_k	Capital depreciation; user cost of capital is $r + \delta_k$
s	Retention rate, the share of total resources rolled into a_{t+1}
z_t	Entrepreneurial productivity in period t
α	Curvature of revenue $R_t = z_t k_t^\alpha$; $\alpha < 1$ delivers DRS in k
λ	Leverage multiplier on net worth: $k_t \leq \lambda a_t$ (equivalently $b_t \leq (\lambda - 1)a_t$)

Three things pinned down: returns to scale (α), user cost (r, δ_k), and tightness of external finance (λ)

The dynamic model adds accumulation: net worth is a state

Each period the entrepreneur inherits the state (a_t, z_t) , then chooses in sequence:

1. **Scale** (static): $k_t \in \arg \max_{k \geq 0} \pi(k, z_t) \quad \text{s.t.} \quad k \leq a_t + b_t, \quad b_t \leq (\lambda - 1) a_t \quad \Rightarrow \quad k_t \leq \lambda a_t$

Cash flow: $\pi(k_t, z_t) = y_t - (r + \delta_k) k_t, \quad y_t = z_t k_t^\alpha$

2. **Saving** (dynamic): $\max_{\{c_t\}} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \log c_t \quad \text{s.t.} \quad \underbrace{c_t + a_{t+1} = (1 + r) a_t + \pi(k_t, z_t)}_{\text{budget}}$

Consumption: $c_t = (1 - s) [(1 + r) a_t + \pi_t], \quad s = \beta$

Net worth: $a_{t+1} = s [(1 + r) a_t + \pi(k_t, z_t)]$

The saving choice drives everything: k_t is purely within-period; choosing c_t is choosing a_{t+1} , tomorrow's collateral

Investment $i_t \equiv k_t - (1 - \delta_k) k_{t-1}$ is accounting only. It becomes a separate *choice* once adjustment costs or irreversibility make k_{t-1} a state

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The Baseline

Self-Financing and Convergence

A patient entrepreneur expands gradually

Deterministic productivity $z_t = \bar{z}$. Budget identity and optimal policy:

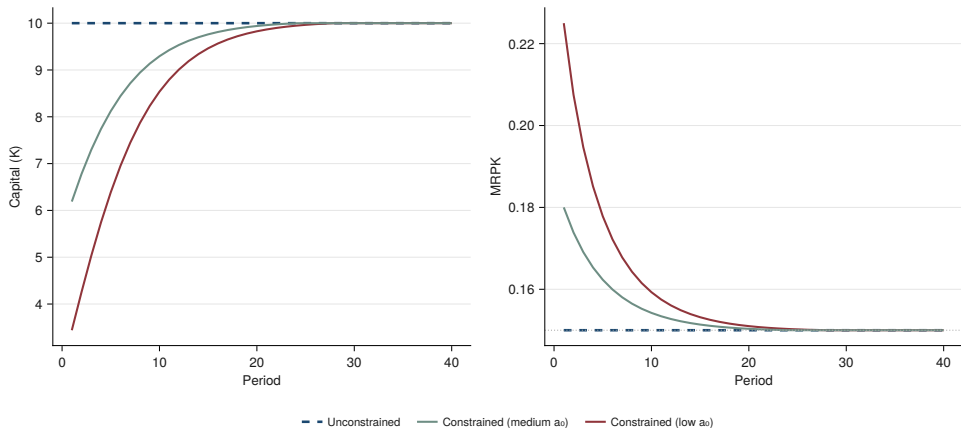
$$y_t = \bar{z}k_t^\alpha, \quad \pi(k_t, \bar{z}) = y_t - (r + \delta_k)k_t, \quad k_t = \min\{k^*(\bar{z}), \lambda a_t\},$$
$$c_t + a_{t+1} = (1 + r)a_t + \pi(k_t, \bar{z}), \quad \underbrace{c_t = (1 - s)[(1 + r)a_t + \pi_t]}_{\text{log-utility policy}},$$

$$a_{t+1} = s[(1 + r)a_t + \pi(k_t, \bar{z})]$$

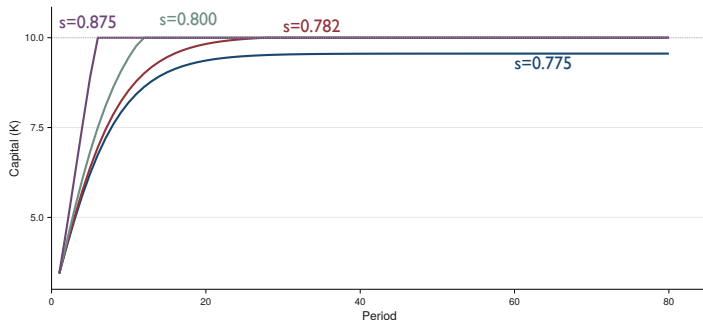
- Low-wealth firm: starts far below k^* , binding for many periods
- Medium-wealth firm: starts closer, binding only briefly
- Each period the entrepreneur consumes $c_t = (1 - s)$ of total resources and retains s . Net worth grows; the constraint relaxes

→ **Key:** does this process actually converge to k^* ?

Self-financing dissolves the static poverty trap



Retention governs convergence speed



Same firm, different s

s	Not retained	Crossing
0.775	22.5%	No crossing by 80
0.782	21.8%	Period 28
0.800	20.0%	Period 12
0.875	12.5%	Period 6

Crossing = first period with $\lambda a_t \geq k^*$

Sharp threshold:

- Just below it, no crossing
- Just above, slow convergence

With enough retention, deterministic frictions are temporary

Key takeaway

Static “poverty traps” dissolve dynamically when productivity is **deterministic** and s is high enough

→ For **persistent wedges** we need something more

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Entry and Exit

Economy-Level Persistence

Individual firms escape; the economy refreshes the pool

Individual law of motion unchanged:

$$k_{i,t} = \min\{k^*(\bar{z}), \lambda a_{i,t}\}, \quad a_{i,t+1} = s[(1+r)a_{i,t} + \pi(k_{i,t}, \bar{z})]$$

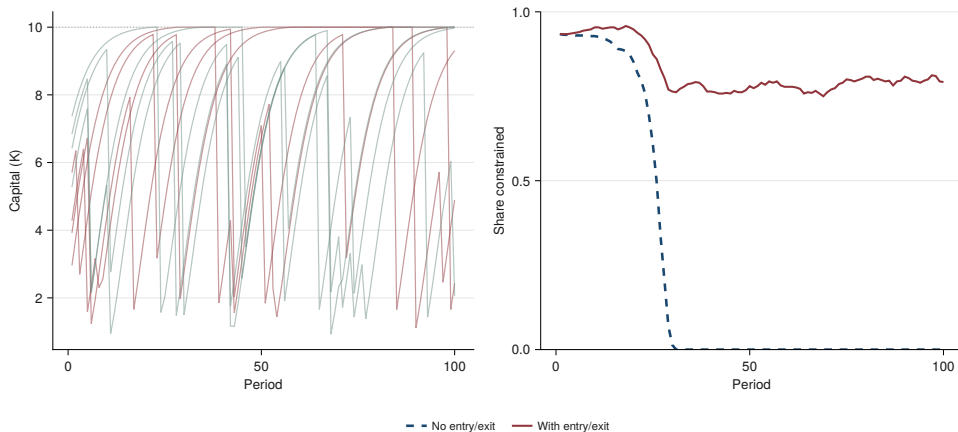
Add demographic turnover:

$$\Pr(\text{exit}_{i,t} = 1) = \delta_e = 0.05, \quad a_{i,t+1} = a_{\text{entrant}} \quad \text{if firm } i \text{ is replaced}$$

Entrants arrive with **low net worth** a_{entrant} , well below the binding threshold k^*/λ . They are constrained on arrival

BKS mechanism: demographic turnover sustains aggregate misallocation even when no single firm is permanently trapped

Demographic turnover sustains economy-level misallocation



Turnover, not traps

Key takeaway

Economy-level persistence does not require firm-level traps

Key distinction: **economy-level** persistence comes from **demographic turnover**, *not* from **firm-level** persistence from **inability to accumulate**

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Stochastic Productivity

Reversible Capital, Moving Target

What if productivity itself is uncertain?

Same collateral and self-financing equations, but **target moves**:

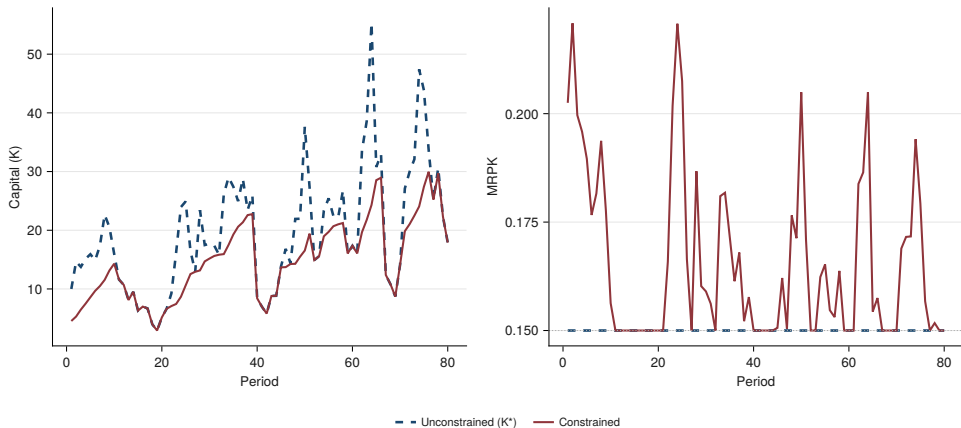
$$y_t = z_t k_t^\alpha, \quad k_t = \min\{k^*(z_t), \lambda a_t\}, \quad a_{t+1} = s[(1+r)a_t + \pi(k_t, z_t)]$$

Idiosyncratic AR(1) productivity:

$$\log z_{t+1} = \log \bar{z} + \rho_z(\log z_t - \log \bar{z}) + \sigma_z \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, 1)$$

→ First-best $k^*(z_t)$ is now a **moving target**. Even a firm at its steady-state wealth can be pushed back into the constraint by a positive z shock that raises k^* above λa

Stochastic productivity creates episodic binding



Persistence of z controls the force of self-financing

The Moll (2014) result:

High persistence ($\rho_z = 0.95$)

Half-life of z : ~ 13.5 periods. Net worth has time to adjust

$\Rightarrow$ Self-financing keeps up. Misallocation stays low

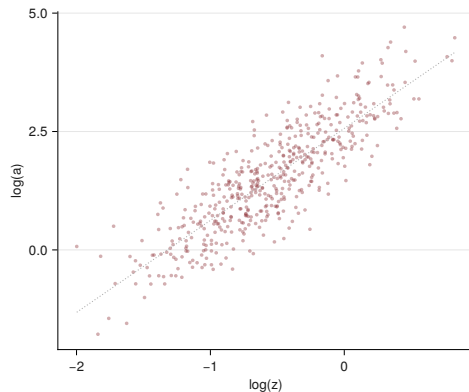
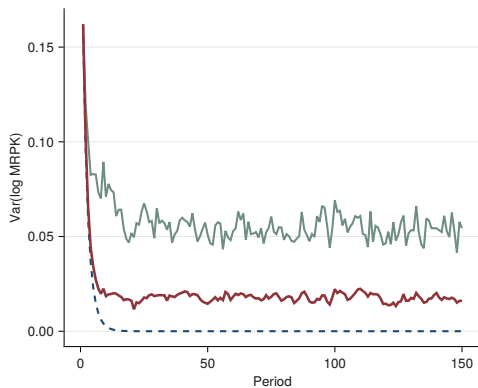
Low persistence ($\rho_z = 0.50$)

Half-life of z : 1 period. The target moves before net worth catches up

$\Rightarrow$ Opportunity vanishes first. Misallocation persists

Innovation size σ_ε rescaled by $\sqrt{1 - \rho_z^2}$ in each case, so both processes have the same long-run dispersion of $\log z$; only persistence differs

Low persistence generates more steady-state misallocation



- - Deterministic — Low persistence ($\rho = 0.5$) — High persistence ($\rho = 0.95$)

Transient opportunities defeat self-financing

With high persistence, high- z firms have time to accumulate toward k^* before z changes

With low persistence, z changes too rapidly for self-financing to keep pace

Opportunity persistence is the key parameter governing whether self-financing resolves or perpetuates frictions

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Useful bells and whistles are not enough

Some standard extensions enrich the model, but do not by themselves generate persistent financial misallocation

- **Cash buffers:** add liquidity, usually make self-financing easier unless targets move faster than wealth
- **Convex adjustment costs:** smooth investment and create real wedges, but not persistent financial wedges by themselves
- **Depreciation, variable labor, taxes, debt maturity:** useful for calibration, not standalone persistence mechanisms

The core question stays: can net worth keep up with the moving target?

Convex adjustment costs smooth investment and create a real wedge

Capital is now a state. Per-period cash flow:

$$\pi(k_t, i_t, z_t) = \underbrace{z_t k_t^\alpha}_{\text{revenue}} - \underbrace{i_t}_{\text{investment}} - \underbrace{C(i_t, k_t)}_{\text{adjustment cost}}, \quad k_{t+1} = (1 - \delta_k)k_t + i_t$$

$C(i, k)$ enters as a deduction from cash flow every period

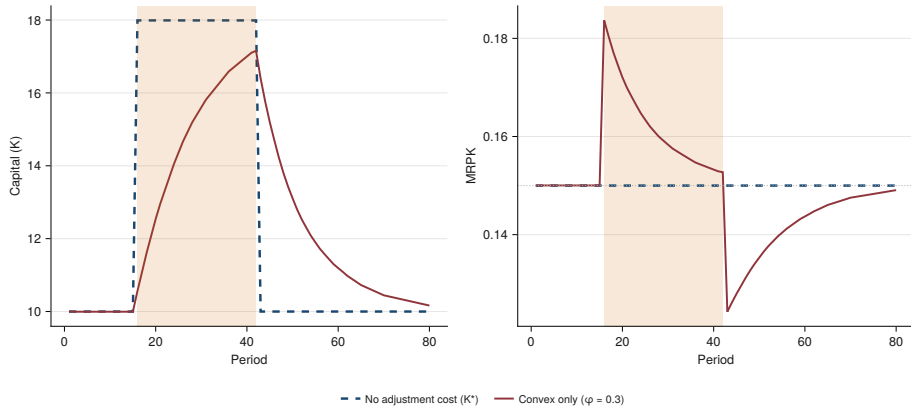
Convex-only specification (Hayashi, q-theory):

$$C(i_t, k_t) = \phi \left(\frac{i_t}{k_t} \right)^2 k_t$$

Behavior. Smooth investment, no inaction region. Firm always adjusts, but by little each period: large bursts are convexly more expensive

Identification point. When $k^*(z_t)$ moves, the firm under-adjusts. Measured MRPK $\neq r + \delta_k$ throughout the smoothing window. **Looks constrained; no collateral constraint binds.** The wedge is technological

Convex case: smooth catch-up, persistent wedge



Yellow band: temporary high-z opportunity. Firm faces only a convex adjustment cost ($\phi = 0.30$). Capital catches up smoothly to $K^*(z)$; MRPK stays above the user cost throughout the smoothing window

Fixed adjustment costs produce lumpy bursts and inaction

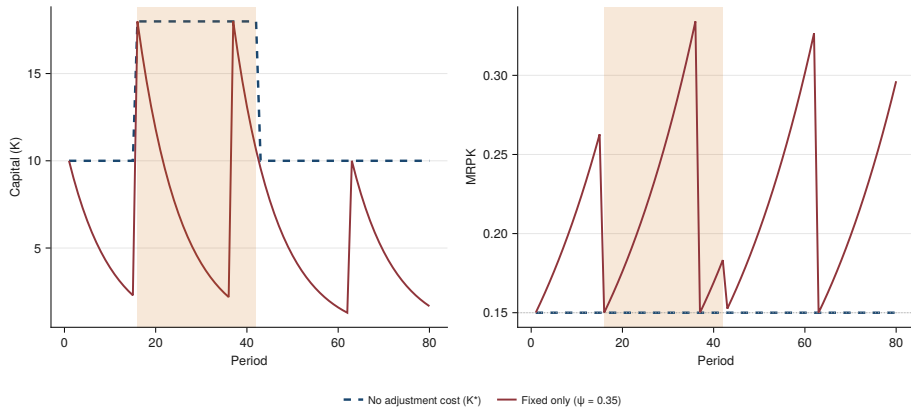
Fixed-only specification (Cooper-Haltiwanger lumpy investment):

$$C_F(i_t, k_t) = \psi \mathbf{1}\left\{\left|\frac{i_t}{k_t}\right| > \vartheta\right\} k_t$$

Behavior. Lumpy or staggered investment. Firm either makes a burst (paying $\psi \cdot k$) or stays in the inaction region with k at passive depreciation

Identification point. When the productivity shock is too small to justify paying ψ , the firm doesn't adjust. k stays put while $k^*(z_t)$ moves. Measured MRPK $\neq r + \delta_k$ throughout the inaction region. Looks constrained; no collateral constraint binds. Same identification problem as convex

Fixed case: lumpy bursts, stale- K wedges



Yellow band: temporary high- z opportunity. Firm faces only a fixed adjustment cost ($\psi = 0.35$). Capital stays at passive depreciation until the gain from adjusting exceeds ψ , then jumps; MRPK spikes during inaction

Combining fixed and convex: the empirical workhorse

Plant-level data show *both* bursty investment AND finite burst size. Neither pure spec matches alone:

- Pure convex predicts perfectly smooth investment
- Pure fixed predicts unbounded jumps when adjusting

Combined form nests both:

$$C(i_t, k_t) = \left[\psi \mathbf{1} \left\{ \left| \frac{i_t}{k_t} \right| > \vartheta \right\} + \phi \left(\frac{i_t}{k_t} \right)^2 \right] k_t$$

Roles. Fixed governs *when* to invest (extensive margin); convex governs *how much* given an adjustment (intensive margin)

Literature. Hayashi (1982), Cooper, Haltiwanger (1993, 1999, 2006), Kehrig, Vincent (2025)

Cross-cutting takeaway. Convex and fixed components, separately or combined, generate measured-MRPK wedges identical to financing constraints. Empirical MRPK dispersion is not by itself evidence of **financing frictions**

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Amplifying the Moving Target

Irreversibility has two faces

Capital resells below its purchase price. After a bad z shock, the firm can:

- Face 1, sell at a loss: downsize anyway, losing χ per unit sold
 - downsizing destroys net worth $\rightarrow$ tighter constraint later (**financial** scarring)
 - MRPK stays *above* $r + \delta_k$: looks like a financing constraint
- Face 2, hold the capital: avoid the loss by not selling
 - keeps $k > k^*(z)$: capital overhang, too much capital for current productivity
 - MRPK falls *below* $r + \delta_k$: a **real** wedge with the opposite sign
 - same inaction logic as fixed costs, but only for downsizing: investing stays free, only selling is penalized (Abel, Eberly 1996; Bertola, Caballero 1994)

$\rightarrow$ Opposite wedge signatures: face 1 mimics a financing constraint, face 2 cannot

The simulation studies face 1: sell at a loss

Start from the stochastic-productivity baseline:

$$a_{t+1} = s[(1+r)a_t + \pi(k_t, z_t)], \quad k_t = \min\{k^*(z_t), \lambda a_t\}$$

Add a resale loss when capital falls:

$$a_{t+1} = s[(1+r)a_t + \pi(k_t, z_t) - \chi \max\{k_{t-1} - k_t, 0\}]$$

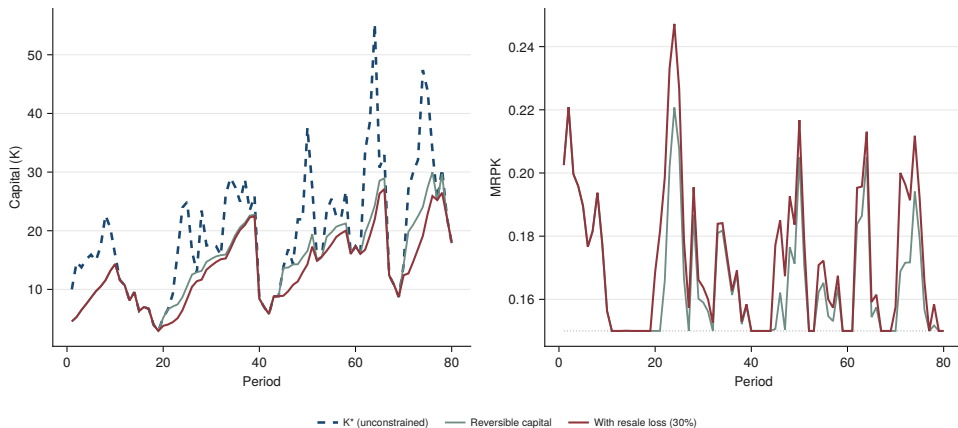
Sunk costs are an amplifier, not a prerequisite: the moving target alone already sustains misallocation

What they add: bad shock $\rightarrow$ firm downsizes, selling capital at a loss (χ per unit) $\rightarrow$ lower $a_{t+1} \rightarrow$ tighter constraint when the next good shock arrives

The policy rule $k_t = \min\{k^*(z_t), \lambda a_t\}$ is deliberately *not* re-optimized: an optimizing firm would sometimes choose face 2 (hold k), so the scarring here is an upper bound

Resale losses amplify and prolong constraint episodes

Same AR(1) shocks. Only change in the environment is the **resale loss** on downsizing



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Paco Buera, Jeremy Majerovitz, Yongseok Shin, Kuldeep Singh, “Risky Business and the Process of Development”

Risk Aversion

When Risk-Bearing Binds

BMSS changes the friction: risk-bearing capacity

Previous sections: capital is a **within-period scale choice**

$$k_t \in \arg \max_{k \geq 0} \pi(k, z_t) \quad \text{s.t.} \quad k \leq a_t + b_t, \quad b_t \leq (\lambda - 1) a_t$$

BMSS is related, but changes the state. Capital is **durable, risky, and partly irreversible**:

$$\text{state } (a_t, k_t, z_t), \quad a_t \geq -\lambda k_t$$

Pedagogical one-period version:

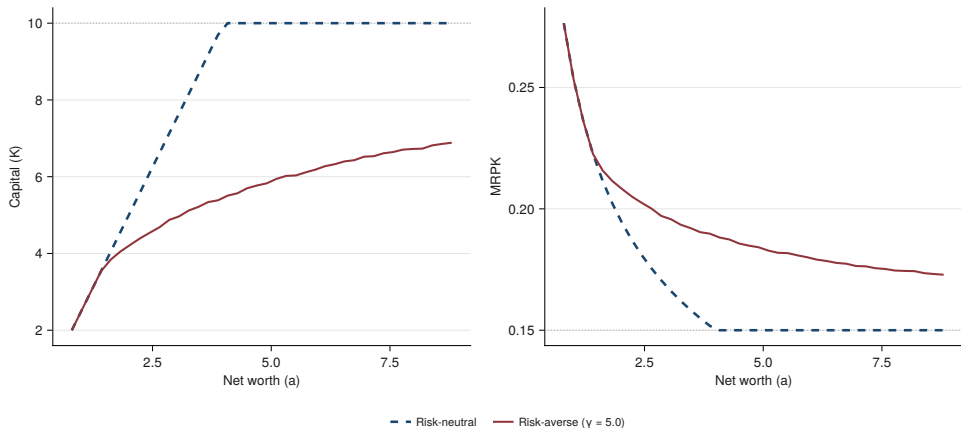
$$\max_k \mathbb{E}[u(W_{t+1})], \quad W_{t+1} = s \cdot [(1+r)a + zk^\alpha - (r + \delta_k)k],$$

where $u(W) = W^{1-\gamma} / (1-\gamma)$

Even when collateral no longer binds, low net worth limits **risk-bearing capacity**, so the entrepreneur may choose $k < k^*$

BMSS (2024): self-insurance is slower than self-financing

Risk-averse entrepreneurs invest below k^* at every wealth level



Risk-bearing is a distinct source of persistent frictions

Risk-neutral MRPK drops to user cost once the firm is wealthy enough

Risk-averse MRPK stays elevated (~ 0.18) across the same wealth range

Self-insurance is an order of magnitude slower than self-financing. The wedge shrinks with wealth, but so slowly that firms never accumulate enough for it to die: it survives in the stationary distribution

This generates **firm-level** persistence, fundamentally different from the collateral channel

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What Have We Learned?

A taxonomy of persistence mechanisms

Mechanism	Persistence level	Key reference
Self-financing (deterministic z)	None (convergence)	Baseline
Entry / exit	Economy-level	Buera, Kaboski, Shin (2011)
Stochastic productivity	Cross-sectional moving target	Moll (2014)
Adjustment costs	Real adjustment wedge	Kehrig, Vincent (2017)
Sunk costs / irreversibility	Amplifier and scarring	—
Risk aversion	Firm-level (permanent)	Buera, Majerovitz, Shin, Singh (2024)

The main persistence mechanisms are [turnover](#), [moving targets](#), and [risk-bearing](#). Adjustment costs and sunk costs mainly shape or amplify the dynamics

The question is not whether firms can self-finance in principle

The question is whether net worth can keep up with the moving target

Appendix

Model parameters

Parameter	Symbol	Value	Role
One-factor curvature	α	0.62	Profit / revenue curvature
Safe rate	r	0.05	Return on safe asset
Depreciation	δ_k	0.10	Capital depreciation
Leverage multiplier	λ	2.50	Constraint $k \leq \lambda a$ (Moll, BKS)
Retention rate	s	0.782	Fraction retained in net worth
Productivity persistence	ρ_z	0.90	AR(1) persistence
Innovation std	σ_z	0.12	Shock size
Exit / resale / risk aversion	δ_e, χ, γ	0.05, 0.30, 5.0	Figures 3, 6, 7
Adjustment costs (convex, fixed)	ϕ, ψ, ϑ	0.30, 0.35, 0.05	Figures 8a, 8b

$\bar{z} = 0.580$ sets $k^*(\bar{z}) = 10$. With $s = 0.782$, steady-state wealth is 0.5% above the collateral threshold

Why the law of motion matters

Naive: $a_{t+1} = a_t + s \cdot \text{income}_t \Rightarrow$ no finite steady state, frictions dissolve mechanically

Correct: $a_{t+1} = s \cdot [(1+r) a_t + \pi_t]$ firm consumes $(1-s)$ of **total resources**, not only of flow profits

Steady state: $a^* = s\pi^* / [1 - s(1+r)]$, exists when $s(1+r) < 1$

With $s = 0.782$: $a^* / \text{threshold} \approx 1.005$. The firm is barely unconstrained at $\bar{z}$, so a positive z shock can push $k^*(z)$ above collateral capacity